

A note on the topology-coincidence statements in
Topology of the space of conormal distributions

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1. Purpose. While using the results of [1] (cited by its published numbering and pages; arXiv:2304.00798v3 has the same numbering, and §§3–4 are unchanged in all arXiv versions), the author found that several statements asserting that two topologies coincide on a whole step of a filtration are false, already in the simplest cases, and that some later proofs depend on them. No erratum could be found (§5). This note is offered to the authors as a courtesy: it lists the statements (§2), gives explicit witnesses with the computation visible (§3), and records what survives and how the affected proofs can be re-routed (§4). The main package of the paper (acyclicity and reactivity of $I, A, \hat{A}, \mathcal{K}, J, K$ for compact M , and the exact sequences) stands once this is done.

2. The statements. The following are false as stated: Prop. 3.2 (p. 16); Cor. 3.4, both assertions (p. 17); the acyclicity and bounded-retractivity clauses of Cor. 3.6 (p. 18) for every non-compact U , together with the sentence on p. 18 extending Props. 3.2–3.3 and Cors. 3.4–3.6 to symbols on bundles over a manifold; Cor. 4.5 (p. 23), for every compact (M, L) with $\text{codim } L \geq 1$; Prop. 6.12 and Cor. 6.14, both assertions (p. 39); Cor. 6.21 (p. 40); Cor. 7.13, both assertions (p. 56). Cor. 6.27 (p. 41) and Cor. 7.22 (p. 58) are true, but their printed proofs pass through Cor. 6.21, hence through Cor. 4.5. The witnesses below are constant in the base variable up to a fixed cut-off, so they also refute every parametrized or bundle version.

3. Witnesses. Notation as in [1]: $S^m = S^m(U \times \mathbb{R}^l)$ with the seminorms $\|\cdot\|_{K,\alpha,\beta,m}$ of (3.1), $\|\cdot\|_{Q,C^k}$ of (3.4), $\|\cdot\|'_{K,\alpha,\beta,m}$ of (3.5); every (3.4) and (3.5) seminorm is continuous on S^m . Fix $\theta \in C_c^\infty(\mathbb{R}^l)$ with $\theta(0) = 1$ and $\text{supp } \theta \subset B(0, r)$.

(a) *Prop. 3.2 and the first assertion of Cor. 3.4.* Let $g_N(x, \xi) = \theta(\xi - Ne_1)$. Then $g_N \in S^{-\infty} \subset S^m$ for every m . If $Q \subset K \times \bar{B}(0, \rho)$, then $\|g_N\|_{Q,C^k} = 0$ for $N > \rho + r$, and every seminorm (3.5) of g_N vanishes because g_N has compact support in ξ . So $g_N \rightarrow 0$ in the topology of (3.4) and (3.5), and $g_N \rightarrow 0$ in $C^\infty(U \times \mathbb{R}^l)$. But for $|\beta| > m$ with $\partial^\beta \theta \not\equiv 0$ (such β exist in every order, a compactly supported nonzero function being no polynomial),

$$\|g_N\|_{K,0,\beta,m} = \sup_{|\eta| \leq r} |\partial^\beta \theta(\eta)| (1 + |\eta + Ne_1|)^{|\beta|-m} \geq (1 + N - r)^{|\beta|-m} \sup |\partial^\beta \theta| \rightarrow \infty.$$

Hence (g_N) is unbounded in S^m : the topology of (3.4) and (3.5) is strictly coarser than that of S^m , for every $m \in \mathbb{R}$, every U and every $l \geq 1$. Taking $|\beta| > m'$: $g_N \in S^m$, $g_N \rightarrow 0$ in C^∞ , $g_N \not\rightarrow 0$ in $S^{m'}$.

(b) *Where the proof of Prop. 3.2 fails.* Let S'^m denote S^m with the topology of (3.4) and (3.5). Fix $R > 2r$, $c_i = (1 + iR)^{m+1}$, and $b_j = \sum_{i \leq j} c_i \theta(\xi - iRe_1)$ (disjoint supports). For $i < j$, $b_j - b_i$ has compact ξ -support inside $\{|\xi| \geq (i+1)R - r\}$, so its (3.5) seminorms vanish and its (3.4) seminorms over a given Q vanish for i large: (b_j) is Cauchy in S'^m . Its C^∞ -limit is the locally finite sum $b = \sum_i c_i \theta(\xi - iRe_1)$, and $|b(x, iRe_1)| (1 + iR)^{-m} = 1 + iR \rightarrow \infty$, so $b \notin S^m$ and (b_j) has no limit in S'^m : S'^m is not complete, contrary to “Hence $S'^m(U \times \mathbb{R}^l)$ is complete” (p. 17). Concretely, for the class a of (b_j) in the completion one has $\|a\|'_{K,0,0,m} = \lim_j \|b_j\|'_{K,0,0,m} = 0$ while $\|\varphi(a)\|'_{K,0,0,m} = \|b\|'_{K,0,0,m} = \infty$; the identity “ $\|\varphi(a)\|'_{K,\alpha,\beta,m} = \|a\|'_{K,\alpha,\beta,m} < \infty$ ” at the top of p. 17 is where the argument breaks.

(c) *Second assertion of Cor. 3.4.* Let $c_j = e^j \theta(\xi - je_1)$, so $c_j \in S^m$ for all m and $c_j \rightarrow 0$ in C^∞ . Fix $x_0 \in U$, $K = \{x_0\}$, and for $k \in \mathbb{N}$ put $\varepsilon_k = \frac{1}{2} \inf_{j \geq 1} e^j (1 + j)^{-k} > 0$ and $W_k = \{a \in S^k : \|a\|_{K,0,0,k} < \varepsilon_k\}$.

*hello.jay.tyagi@gmail.com. Use of AI. The mathematics in this note (the derivations, the computations, and the text) was produced by Claude (Anthropic) working under the author’s direction; the author set the objectives, made the technical and editorial decisions, and is responsible for the content. Claude is a tool and is not an author.

The absolutely convex hull W of $\bigcup_k W_k$ meets every step S^k in a 0-neighborhood, so it is a 0-neighborhood of $S^\infty = \text{ind}_k S^k$. If $c_j = \sum_i \lambda_i w_i$ with $\sum_i |\lambda_i| \leq 1$ and $w_i \in W_{k_i}$, evaluation at (x_0, je_1) gives $e^j = |c_j(x_0, je_1)| < \sum_i |\lambda_i| \varepsilon_{k_i} (1+j)^{k_i} \leq \frac{1}{2} e^j$, a contradiction. So $c_j \notin W$ for all j , and $c_j \not\rightarrow 0$ in S^∞ : the topologies of S^∞ and C^∞ differ on every S^m . (No regularity of S^∞ is used.)

(d) *Cor. 4.5.* Let M be compact, $\text{codim } L = n' \geq 1$, $m < m' < m''$, and $\bar{m}, \bar{m}', \bar{m}''$ as in (4.9). Take $p_0 \in L$, a chart $(U_1, x = (x', x''))$ adapted to L with $x(p_0) = 0$, and the partition of unity in (4.10) with $f_1 = 1$ and $h = f_j = 0$ ($j \geq 2$) near p_0 (replace any $\{h, f_j\}$ by $\{(1-\rho)h, \rho + (1-\rho)f_1, (1-\rho)f_j\}$, $\rho \in C_c^\infty(U_1)$, $\rho = 1$ near p_0 ; by the closed graph theorem the Fréchet topology of $I^m(M, L)$ is independent of these choices anyway). Let $g \in C_c^\infty(\mathbb{R}^{n'})$, $g \not\equiv 0$; $\psi \in C_c^\infty(\{|x'| < 1\})$, $\psi \not\equiv 0$; $N_0 \in \mathbb{N}_0$ with $2N_0 \geq \bar{m}''$; $\psi_* = \Delta^{N_0} \psi$, so $\hat{\psi}_*(\eta) = (-|\eta|^2)^{N_0} \hat{\psi}(\eta)$ is Schwartz and $\partial^\beta \hat{\psi}_*$ vanishes to order $2N_0 - |\beta|$ at 0. Put

$$u_j(x', x'') = j^{n' + \bar{m}'} g(x'') \psi_*(jx') \in C^\infty(M) \subset I^m(M, L).$$

For large j , $hu_j = 0$ and $f_j u_j = 0$ ($j \geq 2$), and by (4.8) the symbol of $f_1 u_j = u_j$ is $a_j(x'', \xi) = j^{\bar{m}'} g(x'') \hat{\psi}_*(\xi/j)$. Substituting $\xi = j\eta$,

$$\|a_j\|_{K, \alpha, \beta, \bar{m}_1} = j^{\bar{m}' - \bar{m}_1} \sup_K |\partial^\alpha g| \Phi_{\beta, \bar{m}_1}(j), \quad \Phi_{\beta, \bar{m}_1}(j) = \sup_\eta |\partial^\beta \hat{\psi}_*(\eta)| (1/j + |\eta|)^{|\beta| - \bar{m}_1}.$$

For $\bar{m}_1 \leq 2N_0$ one has $\Phi_{\beta, \bar{m}_1}(j) \leq C_\beta$ (use the Schwartz decay if $|\beta| \geq \bar{m}_1$, the vanishing at 0 if $|\beta| < \bar{m}_1$), while $\Phi_{0, \bar{m}'}(j) \geq |\hat{\psi}_*(\eta_0)| \min\{(1 + |\eta_0|)^{-\bar{m}'}, |\eta_0|^{-\bar{m}'}\} = c > 0$ for any η_0 with $\hat{\psi}_*(\eta_0) \neq 0$. Hence $\|a_j\|_{K, \alpha, \beta, \bar{m}''} \leq C_j^{\bar{m}' - \bar{m}''} \rightarrow 0$, i.e. $u_j \rightarrow 0$ in $I^{m''}(M, L)$, whereas $\|a_j\|_{K, 0, 0, \bar{m}'} \geq c \sup |g| > 0$, i.e. $u_j \not\rightarrow 0$ in $I^{m'}(M, L)$. The same u_j , with $\text{supp } \psi \subset (\frac{1}{2}, 1)$ in the collar chart of $(\check{M}, \partial M)$, lies in $C_c^\infty(\check{M}) \subset \dot{A}^m(M) = I_M^m(\check{M}, \partial M)$ (6.47) and refutes Cor. 6.21.

(e) *Prop. 6.12, Cor. 6.14, Cor. 7.13.* Let M be compact with boundary, (x, y) a collar chart, $\chi \in C_c^\infty((\frac{1}{2}, 2))$ with $\chi(1) = 1$, $g \in C_c^\infty$ in y with $g(y_0) = 1$, $m' < m$, and $u_j = j^{-m'} \chi(jx) g(y)$. Then $u_j \in C_c^\infty(\check{M}) \subset A^{m_1}(M)$ for every m_1 ; $u_j \rightarrow 0$ in $C^\infty(\check{M})$ (the supports leave every compact set); every seminorm (6.42) of u_j vanishes for j large, and every seminorm (6.43) vanishes for every j ($u_j = 0$ on $\{x < 1/2j\}$). But the seminorm (6.41) with $P = 1$ gives $\sup_{\check{M}} x^{-m'} |u_j| \geq (1/j)^{-m'} j^{-m'} \chi(1) g(y_0) = 1$. So (with $m' = m$) $u_j \rightarrow 0$ in the topology of (6.42) and (6.43) on $A^m(M)$ but not in $A^m(M)$, and (with $m' < m$) $u_j \rightarrow 0$ in $C^\infty(\check{M})$ but not in $A^{m'}(M)$. For the second assertion of Cor. 6.14 take $v_j = e^j \chi(jx) g(y)$ and the 0-neighborhood of $A(M) = \bigcup_k A^{-k}(M)$ built as in (c) from $W_k = \{u \in A^{-k}(M) : \sup x^k |u| < \frac{1}{2} \inf_i e^i i^{-k}\}$; evaluation at $(1/j, y_0)$ shows $v_j \notin W$, so $v_j \not\rightarrow 0$ in $A(M)$ while $v_j \rightarrow 0$ in $C^\infty(\check{M})$. Since $J^m(M, L)$ is given by (7.27) with $\text{Diff}(M, L)$ and $x^m L^\infty(M)$, the same u_j and v_j , placed in $\{\frac{1}{2} < jx < 2\} \subset M \setminus L$, refute both assertions of Cor. 7.13.

(f) *Cor. 3.6 for non-compact U .* Let $U \subset \mathbb{R}^n$ be open and nonempty, $n \geq 1$. Choose $x_j \in U$ leaving every compact subset of U , disjoint closed balls $\bar{B}(x_j, r_j) \subset U$ forming a locally finite family, $\chi_j \in C_c^\infty(B(x_j, r_j))$ with $\chi_j(x_j) = 1$, and $b_j(x, \xi) = \chi_j(x) (1 + |\xi|^2)^{j/2} \in S^j \setminus S^m$ ($m < j$). Then $\{b_j\}$ is bounded in S^∞ : given an absolutely convex 0-neighborhood W , pick in each $W \cap S^k$ a basic neighborhood $W_k = \{a \in S^k : \|a\|_{K, \alpha, \beta, k} < \varepsilon_k, (\alpha, \beta) \in F_k\}$ (F_k finite). For j so large that $\text{supp } \chi_j \cap K_1 = \emptyset$, write $b_j = b_j \rho_R + b_j (1 - \rho_R)$ with $\rho_R(\xi) = \rho(\xi/R)$, $\rho \in C_c^\infty$, $\rho = 1$ near 0. The first term lies in $S^{-\infty} \subset S^1$ and vanishes on $K_1 \times \mathbb{R}^l$, so $2b_j \rho_R \in W_1$; the second lies in S^{j+1} , vanishes for $|\xi| \leq R$, and its S^{j+1} seminorms are $O(1/R)$ by the Leibniz rule, so $2b_j (1 - \rho_R) \in W_{j+1}$ for R large; by absolute convexity $b_j \in W$. The finitely many remaining b_j are absorbed by W . Thus $\{b_j\}$ is bounded in $S^\infty(U \times \mathbb{R}^l)$ but contained in no step: the spectrum is not regular, hence not boundedly retractive and (by [3, Cor. 6.5] as quoted on p. 5) not acyclic; the printed derivations of the other clauses of Cor. 3.6 pass through these properties. The same applies to the extension on p. 18 for bundles over a non-compact manifold, and to [2, §2.1.8, p. 15], which restates Cors. 3.4–3.6 for arbitrary open $U \subset \mathbb{R}^n$. (Sect. 4.3.3, p. 23, excepts acyclicity for non-compact M ; the symbol statements do not.)

4. What survives, and how. For a compact $K \subset U$ let S_K^m be the closed subspace of $S^m(U \times \mathbb{R}^l)$ of symbols vanishing for $x \notin K$, with the seminorms $N_m(a; \gamma) = \|a\|_{K, \alpha, \beta, m}$, $\gamma = (\alpha, \beta)$. For $m < m' < m''$

and every γ there are $N \in \mathbb{N}$, a finite set Γ of indices, and $C, R_0 > 0$ such that

$$N_{m'}(a; \gamma) \leq C \left[R^{-(m'-m)/2} N_m(a; 0) + R^{m''-m'} \max_{\gamma' \in \Gamma} N_{m''}(a; \gamma') \right] \quad (a \in S_K^{m''}, R \geq R_0), \quad (1)$$

proved by Taylor expansion (an inequality of Landau–Kolmogorov type) on boxes of side $\sim (1 + |\xi|)^{-c}$ in x and $(1 + |\xi|)^{1-c}$ in ξ in the region $1 + |\xi| > R$; compact base support makes $N_m(a; 0)$ a global supremum, which is exactly what fails in (f). By (1), the $S_K^{m'}$ - and $S_K^{m''}$ -topologies coincide on the 0-neighborhood $\{N_m(\cdot; 0) < 1\}$ of S_K^m : this is the criterion of [3, Thm. 6.1] quoted on p. 4, so $\bigcup_m S_K^m$ is acyclic. (The inclusions $S_K^m \rightarrow S_K^{m'}$ also map bounded sets to relatively compact sets, by Arzelà–Ascoli and the tail bound $(1 + |\xi|)^{m-m'}$, which is what the Montel clauses need; they are not compact operators, so the remark on p. 5 about compact spectra does not apply as it stands.) The same inequality holds for the collar models of $A^m(M)$, with weight x^{-m} , $x = e^{-\varrho}$, and b -derivatives. Now $I^m(M, L)$ carries by definition the initial topology of (4.10), whose symbols vanish outside the compact base projections of the f_j ; $A^m(M)$ carries the initial topology of $u \mapsto ((\lambda_j u)_j, \mu u)$ into collar models and $C^\infty(M)$, for a partition of unity $\{\lambda_j, \mu\}$ subordinate to collar charts and $\tilde{M}$ (by (6.33) and the local form of $\text{Diff}_b(M)$); $\dot{A}^m(M)$, $\mathcal{K}^m(M)$, $K^m(M, L)$ are closed subspaces of $I^m(\tilde{M}, \partial M)$, $\dot{A}^m(M)$, $I^m(M, L)$ ((6.47), pp. 41, 58); and $J^m(M, L) \cong A^m(M)$ by (7.26). The criterion passes to initial topologies of a fixed map, to finite products with constant spectra, and to subspaces. Hence, for compact M , the LF-spaces $I(M, L)$, $A(M)$, $\dot{A}(M)$, $\mathcal{K}(M)$, $J(M, L)$, $K(M, L)$ are acyclic, hence complete, regular and boundedly, compactly and sequentially retractive by the results of [3] quoted on pp. 4–5. This gives valid proofs of the acyclicity clauses of Cor. 3.6 (compact base, e.g. $S^\infty(\mathbb{R}^l)$), 4.7, 6.16, 6.22, 6.28, 7.15 and 7.23, and then of their Montel clauses.

Two printed arguments use a coincidence statement for more than the word “acyclic”. In the proof of Claim 6.46 (pp. 48–49), the sentence “By Corollary 6.14, there is some 0-neighborhood $V \subset A(M)$ such that $V \cap A^m(M) = W \cap A^m(M)$ ” can be replaced as follows: the absolutely convex hull of A is bounded in some $A^m(M)$, on which the topologies of $A(M)$, $A^{m'}(M)$ and $A^m(M)$ coincide by bounded reactivity, so there is V with $V \cap A \subset W$; this suffices, with $E_{m'}$ from Prop. 6.29 (whose domain is $A^{m'}(M) \supset A^m(M)$, so $m' < m$ rather than $m' > m$ is the index needed). Prop. 8.8 (p. 64) is repaired identically with Cor. 7.31; Cor. 7.13 is not needed there. Cor. 3.5 follows directly: for $a \in S^m$ and $m < m'$, $\|a - \chi(\xi/R)a\|_{K, \alpha, \beta, m'} \leq CR^{m-m'}$ by the Leibniz rule, and a cut-off in x finishes. Finally, Cor. 6.27 and Cor. 7.22 are true: by Prop. 7.26 every element of $K^m(M, L)$ is a finite sum of Dirac layers $\partial_x^k \delta_L \otimes v_k$ with $k \leq \bar{m}$, whose local symbols are polynomials in ξ of degree $\leq \bar{m}$, and on polynomial symbols of bounded degree k_0 every $S^{\bar{m}_1}$ -topology with $\bar{m}_1 \geq k_0$ is the C^∞ -topology of the coefficients; Cor. 6.27 is the case $(\tilde{M}, \partial M)$ via (6.49). In the memoir [2], the uses of compact reactivity of $I(\mathcal{F}) = I(M, M^0; \Lambda \mathcal{F})$ and $J(\mathcal{F})$ (§5.2.1, p. 119; §§5.5.3–5.5.4, p. 122) concern a closed M and are covered by the above.

5. Closing. As of September 3, 2026 the author found no erratum: arXiv lists versions v1 to v3 only; the Crossref record of the DOI carries no correction; zbMATH 7901419 carries no review or corrigendum; Semantic Scholar lists no citing work; Unpaywall lists no correction DOI. Full derivations of (a)–(f), of (1), and of §4 are available on request. The author would be glad to learn whether the authors agree, and thanks them for a paper he has found very useful.

References

- [1] J. A. Álvarez López, Y. A. Kordyukov, E. Leichtnam, Topology of the space of conormal distributions, *J. Pseudo-Differ. Oper. Appl.* 15 (2024), article 47, 68 pp., <https://doi.org/10.1007/s11868-024-00617-y>; arXiv:2304.00798 (v3, 1 June 2024).
- [2] J. A. Álvarez López, Y. A. Kordyukov, E. Leichtnam, A trace formula for foliated flows, arXiv:2402.06671 (v1, 7 February 2024).
- [3] J. Wengenroth, *Derived Functors in Functional Analysis*, Lecture Notes in Mathematics, vol. 1810, Springer, Berlin (2003).